Gyeonghun Kang (姜暻勛)

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Research Interests

Education

Duke University — Durham, NC
Ph.D. Candidate in Statistical Science — Aug. 2023 – Dec. 2026 (expected)
Advisors: Dr. Simon Mak and Dr. Xiang Cheng

Yonsei University — Seoul, South Korea
M.S. in Statistics and Data Science — Mar. 2021 – Feb. 2023
Advisor: Dr. Seonghyun Jeong

Yonsei University — Seoul, South Korea
B.A. in Economics; Minor in Applied Statistics — Mar. 2013 – Feb. 2021

Papers

Selected Work

*†First author; *equal contribution.*

Kang, G.†*, Liu, P.*, Qian, B.*, Cheng, X. (2026), "The Noise-Transmission Burden of Diffusion Network: Why Image Prediction Can Outperform Noise Prediction".

Kang, G.†, Chen, W. X., Cheng, X. (2026), "How Do DiTs Denoise? An Aggregator–Predictor View of Attention and MLPs".

Kang, G.†, Lee, C., Cheng, X. (2026), "Transformers Can Learn Posterior Predictive Distributions In-Context". *International Conference on Machine Learning (ICML)*

Other Work

Kang, G.†, Mak, S., "Bayesian Neural Operators for Uncertainty-Aware Predictions of Expensive Parametric Partial Differential Equations". *In preparation*

Kang, G.†, Mao, J., Ma, L. (2026), "Multiscale Cochran–Mantel–Haenszel Scanning for Conditional Dependency".

Kang, G.†, Jeong, S. (2026), "Model Selection-Based Estimation for Generalized Additive Models Using Mixtures of g-Priors". *Bayesian Analysis*. First published online in 2024.
2022 Brain Korea Excellent Paper Award; 2022 KSS Honorable Mention Poster Award

Honors & Awards

2026 Silver Reviewer Award, *International Conference on Machine Learning (ICML)*
2022 Honorable Mention Poster Award, *Korean Statistical Society Conference*
2022 Excellent Paper Award, *Brain Korea Academic Conference*
2021–22 Brain Korea 21 Scholarship, *National Research Foundation of Korea*
2021 Academic Research Fellowship, *Yonsei University*

Presentations

Posters

2026. Transformers Can Learn Posterior Predictive Distributions In-Context. *International Conference on Machine Learning (ICML)*, Seoul, South Korea

2022. Model Selection-Based Estimation for Generalized Additive Models Using Mixtures of g-Priors. *Korean Statistical Society Conference*, Jeju, South Korea

Talks

2026. Multiscale Cochran–Mantel–Haenszel Test for Conditional Dependency. *Department of Statistical Science*, Durham, NC.

2022. Model Selection-Based Estimation for Generalized Additive Models Using Mixtures of g-Priors. *Korea Data Mining Society Conference*, Busan, South Korea.

Teaching Experience

2026	Teaching Assistant, STA 561: Probabilistic Machine Learning, *Duke University*
2025	Teaching Assistant, STA 610: Multilevel and Hierarchical Models, *Duke University*
2025	Teaching Assistant, STA 332: Theory of Inference, *Duke University*
2024	Teaching Assistant, STA 310: Generalized Linear Models, *Duke University*
2021–22	Teaching Assistant, STA 3127: Statistical Computing, *Yonsei University*

Academic Service

Reviewer: ICML, NeurIPS, ICLR, and AAAI (2026).

Languages

Korean (native); English (fluent); Mandarin Chinese (conversational; currently studying).